DCF Model

Assumptions	
Perpetural Growth Rate	2%
Transaction Date	*31/12/2026*
Fiscal Year End	31/12/2026
Current Price	39.00
Shares Outstanding	1,165
Risk Free Rate	2.5%
Market Return	8.5%

Competitors	Levered B	Gearing
Competitor A	1.30	40%
Competitor B	1.60	55%
Competitor C	1.50	50%

	(using Hamac
	Unlevered B
Means that D is 40% and E is 60%	1.02
Means that D is 55% and E is 45%	1.16
Means tht D is 50% and E is 50%	1.11
Average	**1.09**

Q) Find the value of the company.

Ans:

Step 1: Find the Tax rate of each y in IS using the formula:

$$\text{Tax Rate} = \frac{\text{Income Tax}}{\text{Profit Before Tax}}$$

Step 2: Take the tax rate of 2025, since here the transaction date is 2025.

Step 3: Find the Unlevered Beta using the Hamada Equation.

✅ **Hamada Equation:**

$$\beta_L = \beta_U \left(1 + (1 - T) \cdot \frac{D}{E}\right)$$

Rearranged to solve for **Unlevered Beta** β_U:

$$\beta_U = \frac{\beta_L}{1 + (1 - T) \cdot \frac{D}{E}}$$

Where:

- β_L = Levered Beta (Equity Beta)
- β_U = Unlevered Beta (Asset Beta)
- D/E = Debt-to-Equity ratio (from gearing)
- T = Tax rate

Step 4: Find the average of the three Unlevered Beta.

Step 5: Find the D/E ratio in BS for all years.

Step 6: Re-lever the average Unlevered beta, found in Step 4, using the D/E ratio of our company in 2025.
This is your company's levered Beta now.

Step 7: CAPM Calculation for finding Re

$$r = R_f + \beta(R_m - R_f)$$

Step 8: Write the Re from CAPM, the interest from 2025, the debt from 2025, and the calculated Rd.

Step 9: Calculate WACC

$$\text{WACC} = \left(\frac{E}{D+E}\right) \cdot R_e + \left(\frac{D}{D+E}\right) \cdot R_d \cdot (1-T)$$

Step 10: Create the Discounted CFs table.

Step 11: Create a new sheet to calculate Capex and then WCN

Step 12: Calculate the FCFs.

Free Cash Flow (FCF) Formula (Unlevered FCF):

$$\text{FCF} = \text{EBIT} \cdot (1-T) + \text{Depreciation} - \text{CapEx} - \Delta\text{WCN}$$

Step 13: Calulate using DCF model.

DCF Valuation Formula:

$$\text{Value} = \sum_{t=1}^{n} \frac{FCF_t}{(1+WACC)^t} + \frac{TV}{(1+WACC)^n}$$

Terminal Value (TV) Formula — Gordon Growth Model:

$$\text{TV} = \frac{FCF_{n+1}}{WACC - g}$$

$$FCF_{2031} = FCF_{2030} \times (1+g)$$

Step 14: Sum the FCFs and TV to get the EV

Step 15: Calculate the Equity Value

Step 16: Calculate the Equity value per shares outstanding

da Eq)

eta

Here, we have levered debt. However, in cases where we have no debt, then it will be unlevered beta.
For only unlevered beta: 1) Unlever all beta 2) Find their average
For with levered beta: 1) Use Hamada Equation 2) You'll find your replaced beta
Levered beta is the equity beta.
D/E ratio is one form of gearing ratio.

We can't take the average of l
Instead, we unlever their betas
Re-levering the average Unlev

Company Beta: 1.2

evered betas because each company has different debt levels.
s, average the unlevered betas, and then re-lever using our unlevered beta and our projected debt-to-equity ratio for 2025
ered beta calculated using our company's capital structure of 2025, as the transation y given here is 2025.

CAPM:	**9.7%**	
So, Re:	9.7%	
Interest:	3.5	
Debt:	86.6	
Rd:	4%	(i% is interest/debt)
Equity:	495.8	85%
Debt:	86.6	15%
Total:	**582.42**	**100%**
WACC:	**8.678%**	

Discounted CFs	Entry	2025	2026e	2027e	2028e	2029e	2030e
Date	31.12.2025	31.12.2025	31.12.2026	31.12.2027	31.12.2028	31.12.2029	31.12.2030
Time Periods			1	2	3	4	5
EBIT (income from operations)			848	1,616	2,873	4,755	7,316
Tax Rate			30%	30%	30%	30%	30%
(+) D&A			101	172	276	417	590
(-) Capex			232	406	583	1,014	1,556
(-) Change in WCN			58	317	271	366	450
(=) FCF			**405**	**579**	**1,432**	**2,364**	**3,705**

DCF Model	2025	2026e
Discounted FCF		373

Terminal Value	**56,585**

Discounted TV	**37,324**

Enterprise Value:	**€**	**43,441**
(+) Cash		376
(-) Debt		70
(=) Equity Value	**€**	**43,748**
Equity Value/Share	**€**	**37.55**

(share price is overvalued)

2027e	2028e	2029e	2030e	Total
490	1,115	1,695	2,444	**6,117**